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The Dichromatic Number of a Digraph

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In this paper the concept of *dichromatic number of a digraph* which is a generalization of the chromatic number of a graph is introduced. The *dichromatic number* of a digraph D is defined as the minimum number of colours required to colour the vertices of D in such a way that the chromatic classes induce acyclic subdigraphs in D. Some results relating the dichromatic number of D with the existence of cycles of special lengths in D are presented. Contributions to chromatic theory are also obtained. In particular, we generalize the theorem due to P. Erdös and A. Hajnal (*Acta Math. Acad. Sci. Hungar.* **17** (1966), 61–99) which states the existence of odd cycles of length $\geqslant \chi(G) - 1$ in any graph G.

TERMINOLOGY

Let D be a digraph, $V(D)$ and $A(D)$ will denote the sets of vertices and arcs, respectively. By walks, paths, and cycles we mean directed walks, paths, and cycles. If G is a graph, G^* is the digraph defined by

(i) $V(G^*) = V(G)$,

(ii) $(u, v) \in A(G^*)$ iff $[u, v] \in E(G)$.

Notice that for each $[u, v]$ in $E(G)$ there are two arcs (u, v) and (v, u) in $A(G^*)$.

We shall denote the set $\{0, 1,..., n-1\}$ by I_n. For general terminology we refer the reader to Harary [5]. In particular $\chi(G)$ denotes the chromatic number of G.

DEFINITION 1. An *acyclic n-colouring* of a digraph D is a function $f\colon V(D) \to I_n$ such that $f^{-1}(i)$ induces an acyclic subdigraph in D for each $i \in I_n$. The *dichromatic number* $d_k(D)$ of D is the minimum n such that there exists an acyclic n-colouring of D. D is said to be *n-dichromatic* if $d_k(D) = n$

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and *minimal n-dichromatic* if $d_k(D) = n$ and $d_k(D_0) < n$ for every proper subdigraph D_0 of D.

Remark 1. (i) $d_k(D_1) \leqslant d_k(D_2)$ whenever $D_1 \subseteq D_2$.

(ii) If $d_k(D) \geqslant n$, D contains a minimal n-dichromatic subdigraph.

(iii) If D is minimal $(n+1)$-dichromatic and $(u, v) \in A(D)$, then $d_k(D-(u, v)) = n$ and $f(u) = f(v)$ for every acyclic n-colouring of $D-(u, v)$.

Remark 2. $d_k(G^*) = \chi(G)$. Moreover, if G is minimal n-chromatic, then G^* is minimal n-dichromatic.

THEOREM 1. *If D is minimal n-dichromatic, D is strongly connected and has no cutpoints.*

Proof. Suppose D is not strongly connected. Each strong component of D can be coloured with colours in I_{n-1} since D is minimal n-dichromatic. The join of the partial colourings gives an acyclic $(n-1)$-colouring of D contradicting the hypothesis. It is obvious that D has no cutpoints.

COROLLARY 1.

$$d_k(D) = \max\{d_k(H) \mid H \textit{ is strong component of } D\}.$$

As an application of Corollary 1 we shall prove that if D does not contain odd cycles, then $d_k(D) \leqslant 2$.

If D is strongly connected and does not contain odd cycles, D is bipartite [6, Theorem 6.14]. The general case follows from Corollary 1.

DEFINITION 2. Let $f: V(D) \to I_n$ be an acyclic n-colouring of D and p any permutation of I_n. A subset S of $V(D)$ is called *p-stable* (w.r.t. f) iff $(u, v) \in A(D)$, $u \in S$ and $pf(u) = f(v)$ implies $v \in S$. A walk (resp. path) $(u_0, u_1, \ldots, u_m)$ in D is a *p-walk* (resp. *p-path*) iff $pf(u_i) = f(u_{i+1})$ for $i = 0, 1, \ldots, m-1$. A subdigraph D_0 of D is a *p-subdigraph* iff $(u, v) \in A(D_0)$ implies $pf(u) = f(v)$.

Remark 3. S is p-stable in D iff $V(D) - S$ is p^{-1}-stable in D' (D' is the *converse* of D and is obtained from D by reversing the arcs of D).

Remark 4. The intersection of any family of p-stable subsets of $V(D)$ is p-stable.

Let U^p denote the p-stable closure of U, i.e., the unique minimal p-stable subset of $V(D)$ containing U (U^p does exist because of Remark 4 and the fact that $V(D)$ is p-stable).

The proof of Lemma 1 is left to the reader.

LEMMA 1. *U^p is the set of vertices $w \in V(D)$ such that there exists a p-path from U to w.*

BASIC LEMMA. *Let $f\colon V(D) \to I_n$ be an acyclic n-colouring of D and S any p-stable subset of $V(D)$. Then the function $f_1\colon V(D) \to I_n$ defined by*

(i) $f_1 \mid S = p(f \mid S)$

(ii) $f_1 \mid V(D) - S = f \mid V(D) - S$

is also an acyclic n-colouring of D.

Proof. Suppose f_1 is not an acyclic n-colouring of D. Then there exists a cycle $C = (x_0, x_1, \ldots, x_{m-1}, x_0)$ in D such that f_1 is constant in $V(C)$. Clearly, $V(C) \not\subseteq S$ and $V(C) \not\subseteq V(D) - S$ for otherwise f would be constant in $V(C)$. Let k be such that $x_k \in S$ and $x_{k+1} \in V(D) - S$ ($k+1$ is taken mod m). We have $pf(x_k) = f_1(x_k) = f_1(x_{k+1}) = f(x_{k+1})$. Since S is p-stable, $x_{k+1} \in S$, which gives a contradiction.

THEOREM 2. *Let D be a n-dichromatic digraph; $u, v \in V(D)$. Suppose that for every acyclic n-colouring g of D, $g(u) = g(v)$. Let f be an acyclic n-colouring of D and p a permutation of I_n which does not fix $f(u)$. Then there exists a p-path in D from u to v.*

Proof. Vertex v must belong to $\{u\}^p$ for otherwise, by the Basic Lemma, there would be an acyclic n-colouring f_1 of D such that $f_1(u) \neq f_1(v)$. By Lemma 1 there exists a p-path from u to v.

DEFINITION 3. Let D be a digraph, $f\colon V(D) \to I_n$ an acyclic n-colouring of D. A subdigraph D_0 of D is said to be a *C_m-subdigraph of D* iff there exists a cyclic permutation $p = (c_0, c_1, \ldots, c_{m-1})$ such that D_0 is a p-subdigraph of D and $f(V(D_0)) \subseteq \{c_0, c_1, \ldots, c_{m-1}\} \subseteq I_n$.

Remark 5. If D_0 is a C_m-subdigraph of D, every cycle in D_0 has length $\equiv 0 \pmod m$.

Let $c_0(s, m)$ denote the maximum number of edge disjoint cycles of length m in K_s passing by a given vertex. Define $c(s, m) = 2c_0(s, m)$ for $2 < m \leqslant s$ and $c(s, 2) = c_0(s, 2) = s - 1$. Clearly,

$$\left[\frac{s-1}{m-1}\right]\left[\frac{m-1}{2}\right] \leqslant c_0(s, m) \leqslant \left[\frac{s-1}{2}\right].$$

THEOREM 3. *Let D be a digraph, $(u, v) \in A(D)$. If $d_k(D) = k+1$, $d_k(D - (u, v)) = k$, $k \geqslant 2$, and f is an acyclic k-colouring of $D - (u, v)$, then for every m, $2 \leqslant m \leqslant k$, there exist $c(k, m)$ mutually arc-disjoint strongly connected C_m-subdigraphs of $D - (u, v)$ joining u and v.*

Proof. By Remark 1(iii), $f(u) = f(v)$. We can assume that $f(u) = 0$. Call two permutations p and p' independent iff $p(i) = p'(i)$ implies $p(i) = i$. Construct a set $W = \{(0, c_1^j, ..., c_{m-1}^j) \mid j \in J\}$ of $c(k, m)$ mutually independent cyclic permutations of I_k as follows: Consider the complete graph $K(I_k)$ whose vertex set is I_k. If $m = 2$, take $W = \{p_i = (0, i) \mid i = 1, ..., k-1\}$. If $m > 2$ take a set of $c_0(k, m)$ edge-disjoint cycles of length m in $K(I_k)$ passing by 0 and give to each of them the two possible orientations to get two cyclic permutations. Set W is the set of cyclic permutations obtained this way. By Theorem 2, there exist, for each $p \in W$, p-paths P_p and P'_p from u to v and v to u, respectively. Since p fixes no colour occurring in $P_p \cup P'_p$ and W is a set of mutually independent permutations, it follows that the digraphs $P_p \cup P'_p$ form the required set of C_m-subdigraphs of $D - (u, v)$.

Remark 6. If D_0 is one of the C_m-subdigraphs in Theorem 3, every uv-path in D_0 has length $\equiv 0 \pmod m$.

As a direct consequence of Theorem 3 and Remarks 5 and 6 we obtain

THEOREM 4. *Let D be a minimal $(k+1)$-dichromatic digraph, $k \geqslant 2$ and m any integer such that $2 \leqslant m \leqslant k$. Then*

(i) *For any two adjacent vertices u and v in D, there exists a set of $c(k, m)$ mutually arc-disjoint uv-paths of length* $\equiv 0 \pmod m$.

(ii) *Any $(u, v) \in A(D)$ is contained in $c(k, m)$ cycles of length* $\equiv 1 \pmod m$ *satisfying that any two of them have only one common arc, just (u, v).*

(iii) *Every $u \in V(D)$ is contained in $c(k, m)$ arc-disjoint cycles of length* $\equiv 0 \pmod m$.

COROLLARY 2. *If D is a minimal $(k+1)$-dichromatic digraph, $k \geqslant 2$ every $(u, v) \in A(D)$ belongs to an odd cycle of length greater or equal than k.*

Proof. Let us choose $m = 2[\frac{1}{2}k]$.

Corollary 2 generalizes the following result due to Erdös and Hajnal [3].

THEOREM (Erdös and Hajnal [3]). *If $L \geqslant 3$ is the length of the longest odd cycle in G, then $\chi(G) \leqslant L + 1$.*

THEOREM 5. *Let D be a minimal $(k+1)$-dichromatic digraph. Then D is strongly k-line connected.*

Proof. If $k = 1$, the result follows from Theorem 1. If $k \geqslant 2$, take $(u, v) \in A(D)$ and let f be an acyclic k-colouring of $D - (u, v)$. Without loss of generality we can assume that $f(u) = 0$. Consider the permutations $p_i = (0, i)$, $i = 1, ..., k-1$. By Theorem 2 there exist p_i-paths P_i from u to v. These paths and (u, v) give a set of k edge-disjoint uv-paths. Therefore, $\lambda(u, v) \geqslant k$. Let (V_1, V_2) be a partition of $V(D)$. The set $[V_1, V_2]$ of $V_1 V_2$-arcs in D is nonempty because D is strongly connected (Theorem 1). Let $(v_1, v_2) \in [V_1, V_2]$. Since $\lambda(v_1, v_2) \geqslant k$, $|[V_1, V_2]| \geqslant k$ and the proof is complete.

As a corollary we obtain the following result due to Dirac (see [1, Theorem 1]).

COROLLARY 3 (Dirac [1]). *Let G be a minimal $(k+1)$-chromatic graph. Then G is k-line connected.*

Denote by Δ_0 and Δ_i the maximum outdegree and the maximum indegree in D, respectively.

COROLLARY 4. $d_k(D) \leqslant \min\{\Delta_0, \Delta_i\} + 1$.

Theorems 6 and 7 are undirected variations of Theorems 3 and 4.

THEOREM 6. *Let G be a graph, $[u, v] \in E(G)$. If $\chi(G) = k + 1$, $\chi(G - [u, v]) = k$, $k \geqslant 2$, and f is a k-colouring of $G - [u, v]$, then*

(i) *For every m, $2 < m \leqslant k$, there exists a set $\{G_p \mid p \in W\}$ of $c_0(k, m)$ edge-disjoint subgraphs of G joining u and v and a strongly connected orientation $\mathbf{G}_p$ of G_p for each $p \in W$ such that $\mathbf{G}_p$ is a C_m-subdigraph of $G^* - (u, v)$.*

(ii) *For $m = 2$ there exists a set $\{T_p \mid p \in W\}$ of $c_0(k, m)$ edge-disjoint uv-paths such that each T_p oriented from u to v is a C_2-subdigraph of $G^* - (u, v)$.*

Proof. First of all take $D = G^*$ and notice that f is an acyclic k-colouring of $G^* - (u, v)$. Proceed now as in proving Theorem 3 but take only one orientation and then only one cyclic permutation for each of the $c_0(k, m)$ edge-disjoint cycles of length m in $K(I_k)$. Let W be the set of these permutations and continue with the same proof. The case $m = 2$ can also be treated as in the proof of Theorem 3.

Theorem 7 is a direct consequence of Theorem 6.

THEOREM 7. *Let G be a minimal $(k+1)$-chromatic graph, $k \geqslant 2$. Then*

(i) *If $2 \leqslant m \leqslant k$, for any two adjacent vertices u and v in G, there exists a set of $c_0(k, m)$ mutually edge-disjoint uv-paths of length $\equiv 0 \pmod m$.*

(ii) *If $2 \leqslant m \leqslant k$, any $[u, v] \in E(D)$ is contained in $c_0(k, m)$ cycles of length $\equiv 1 \pmod m$ satisfying that any two of them have only one common edge, i.e., $[u, v]$.*

(iii) *If $2 < m \leqslant k$, every $u \in V(G)$ is contained in $c_0(k, m)$ edge-disjoint cycles of length $\equiv 0 \pmod m$.*

Remark 7. Theorem 7(iii) can be completed as follows: If $m = 2$, every $u \in V(G)$ is contained in $[\frac{1}{2}(k-1)]$ edge-disjoint cycles of even length. To see it, take paths P_i as in the proof of Theorem 5 and observe that the union of any two of them contains an even cycle passing by u.

Further applications of the method employed here will be presented in another paper [7].

Theorem 8 gives a relation between $d_k(D)$, $d_k(H)$, and $d_k(D[H])$, where $D[H]$ denotes the *composition* or *lexicographical product* of D and H. (Product $D[H]$ is defined by (i) $V(D[H]) = V(D) \times V(H)$, and (ii) $((u_1, h_1), (u_2, h_2)) \in A(D[H])$ iff $(u_1, u_2) \in A(D)$ or $u_1 = u_2$ and $(h_1, h_2) \in A(H)$.)

THEOREM 8. $d_k(D[H]) \geqslant d_k(D) + d_k(H) - 1$.

Proof. If $d_k(H) = 1$ equality clearly holds. Suppose that $d_k(H) \geqslant 2$. Denote by H_v the isomorphic copy of H induced by $\{v\} \times H$ in $D[H]$. Put $n = d_k(D[H])$, $m = d_k(H) - 1$. Clearly, $n - m \geqslant 1$. Let f: $V(D[H]) \to I_n$ an acyclic n-colouring of $D[H]$. Then $f^{-1}(I_n - I_m) \cap V(H_v) \neq \varnothing$ for every $v \in V(D)$ since $f^{-1}(I_n - I_m)$ misses only $m = d_k(H) - 1$ chromatic classes. Take $w_v \in f^{-1}(I_n - I_m) \cap V(H_v)$. The subdigraph of $D[H]$ spanned by $\{w_v \mid v \in V(D)\}$ is an isomorphic copy of D. This implies $n - m = |I_n - I_m| \geqslant d_k(D)$ and the required inequality follows.

In a forthcoming paper Erdös [4] and the present author introduce and investigate the concept of dichromatic number $d_k(G)$ of a graph G, where

$$d_k(G) = \max\{d_k(G_*) \mid G_* \text{ is an orientation of } G\}.$$

(See also [2].)

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